

Math 472: Homework 06

Due Wednesday, March 11

Problem 1 (Exercise 8.50). Refer to example 8.8 in the textbook. In this example, p_1 and p_2 were used to denote the proportions of refrigerator of brands A and B , respectively, that failed during the guarantee periods.

- At the approximate 98% confidence level, what is the largest “believable value” for the difference in the proportions of failures for refrigerators of brands A and B ?
- At the approximate 98% confidence level, what is the smallest “believable value” for the difference in the proportions of failures for refrigerators of brands A and B ?
- If $p_1 - p_2$ actually equals 0.2251, which brand has the larger proportion of failures during the warranty period? How much larger?
- If $p_1 - p_2$ actually equals -0.1451 , which brand has the larger proportion of failures during the warranty period? How much larger?
- As observed in Example 8.8, zero is a believable value of the difference. Would you conclude that there is evidence of a difference in the proportions of failures (within the warranty period) for the two brands of refrigerators? Why?

Problem 2 (Exercise 8.70). Let Y be a binomial random variable with parameter p . Find the sample size necessary to estimate p within .05 with probability .95 in the following situations:

- If p is thought to be approximately .9
- If no information is known about p (use $p = .5$ in estimating the variance of $\hat{p}$).

Problem 3 (Exercise 8.81). The carapace lengths (in mm) of ten lobsters examined in a study of the infestation of the *Thenus orientalis* lobster by two types of barnacles, *Octolasmis tridens* and *Octolasmis lowei*, are as follows:

78 66 65 63 60 60 58 56 52 50.

Find a 95% confidence interval for the mean carapace length (in mm) of *Thenus orientalis* lobsters.

Problem 4 (Exercise 8.85). Two new drugs were given to patients with hypertension. The first drug lowered the blood pressure of 16 patients an average of 11 points, with a standard deviation of 6 points. The second drug lowered the blood pressure of 20 other patients an average of 12 points, with a standard deviation of 8 points. Determine a 95% confidence interval for the difference in the mean reductions in blood pressure, assuming that the measurements are normally distributed with equal variances.

Problem 5 (Exercise 9.1). Suppose that Y_1, Y_2, Y_3 are a random sample from an exponential distribution with density function

$$f(y) = \frac{1}{\theta} e^{-y/\theta} \mathbf{1}_{[y>0]}$$

Consider the following estimators

$$\hat{\theta}_1 = Y_1, \quad \hat{\theta}_2 = \frac{Y_1 + Y_2}{2}, \quad \hat{\theta}_3 = \frac{Y_1 + 2Y_2}{3}, \quad \hat{\theta}_4 = \min(Y_1, Y_2, Y_3), \quad \hat{\theta}_5 = \bar{Y}$$

In problem #8 of homework 4, you showed that $\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3$ and $\hat{\theta}_5$ are unbiased estimators of θ .

- Find $\text{eff}(\hat{\theta}_1, \hat{\theta}_5)$.
- Find $\text{eff}(\hat{\theta}_2, \hat{\theta}_5)$.
- Find $\text{eff}(\hat{\theta}_3, \hat{\theta}_5)$.

Problem 6 (Exercise 9.6). Suppose $Y_1, \dots, Y_n$ is a random sample of size n from a Poisson distribution with mean λ . Consider $\hat{\lambda}_1 = \frac{Y_1 + Y_2}{2}$ and $\hat{\lambda}_2 = \bar{Y}$. Derive the efficiency of $\hat{\lambda}_1$ relative to $\hat{\lambda}_2$.

Problem 7 (Exercise 9.8). Let $Y_1, \dots, Y_n$ denote a random sample from a probability density function $f(y)$, which has unknown parameter θ . If $\hat{\theta}$ is an unbiased estimator of θ , then under very general conditions

$$\text{Var}(\hat{\theta}) \geq \frac{1}{\mathcal{I}(\theta)}$$

where

$$\mathcal{I}(\theta) = n\mathbb{E} \left[-\frac{\partial^2}{\partial \theta^2} \left[\log(f(Y)) \right] \right].$$

(This is known as the Cramer-Rao inequality.) If $\text{Var}(\hat{\theta}) = \frac{1}{\mathcal{I}(\theta)}$, the estimator $\hat{\theta}$ is said to be *efficient*.

Suppose that $f(y)$ is the normal density with mean μ and variance σ^2 . Show that $\bar{Y}$ is an efficient estimator of μ .

Problem 8 (Exercise 9.17). Suppose that $X_1, \dots, X_n$ and $Y_1, \dots, Y_n$ are independent random samples from populations with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2 respectively. Show that $\bar{X} - \bar{Y}$ is a consistent estimator of $\mu_1 - \mu_2$.

Problem 9 (Exercise 9.24). Let $Y_1, \dots, Y_n$ be independent standard normal random variables.

(a) What is the distribution of $\sum_{i=1}^n Y_i^{2n}$?

(b) Let $W_n = \frac{1}{n} \sum_{i=1}^n Y_i^2$. Does W_n converge in probability to some constant? If so, what is the value of the constant?

Problem 10 (Exercise 9.39). Let $X_1, \dots, X_n$ denote a random sample from a Poisson distribution with parameter λ . Show that $\sum_{i=1}^n Y_i$ is sufficient for λ .